

L02: The Mathematical Building Blocks of Neural Networks

CSci 560 Deep Learning w/ Python (Chollet) Ch. 2

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L02.1 A First Look at a Neural Network

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L02.2 Data Representations for Neural Networks

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L02.3 The Gears of Neural Networks: Tensor Operations

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Matrix · Matrix Dot-Product

- x , y are input matrices of shape (a, b) and (b, c) respectively
- Each row of x performs an inner product with corresponding column of y
- So resulting element $z[i, j]$ is the inner product between row i of x and column j of y
- Resulting shape of z is (a, c)

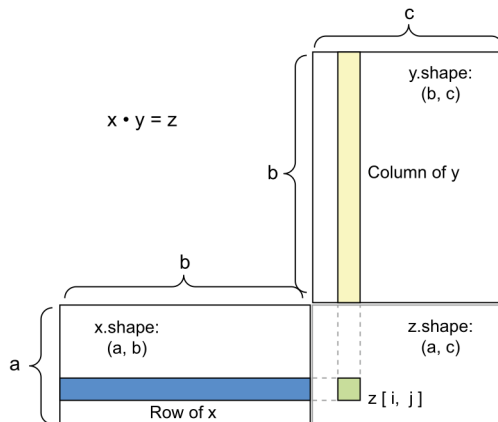


Figure 1: Matrix dot-product box diagram. (Chollet, 2021, pg.43)



Vector Addition is Translation

- Adding a vector to a point will move the point by a fixed amount.
- Applied to a set of points, result is a translation.

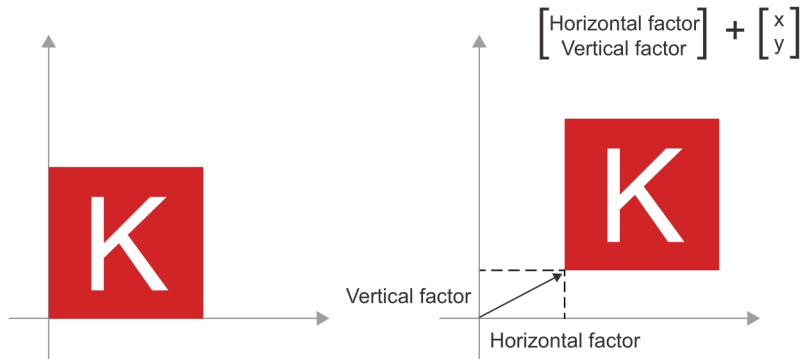


Figure 2: 2D translation as vector addition. (Chollet, [2021](#), pg.45)



Matrix Multiplication (dot product) is Rotation

- A counterclockwise rotation by angle θ achieved by dot product.

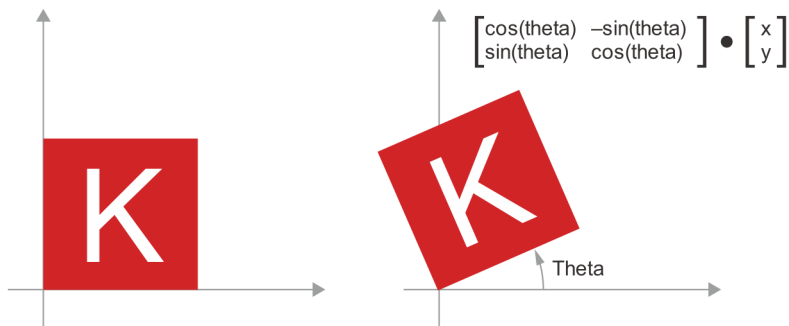


Figure 3: 2D rotation (counterclockwise) as a dot product. (Chollet, 2021, pg.45)

Matrix Multiplication can also Scale

- Vertical and horizontal scaling also achieved by dot product with a suitable matrix.

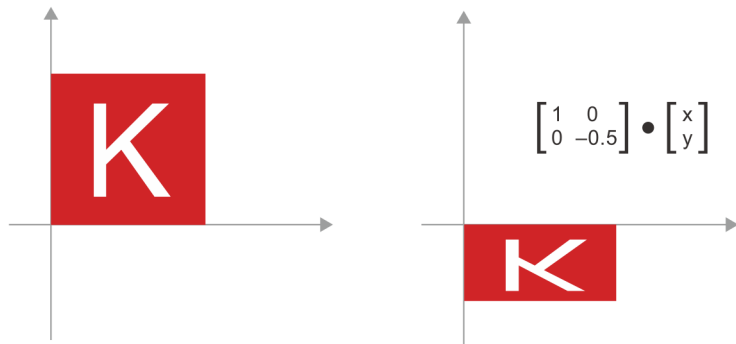


Figure 4: 2D scaling as a dot product. (Chollet, [2021](#), pg.46)

Affine Transformation: Linear transform and translation

- Translation, Rotation, Scaling are all linear transformations.
- Can combine, for example $y = W \cdot x + b$ is a linear transformation plus a translation.

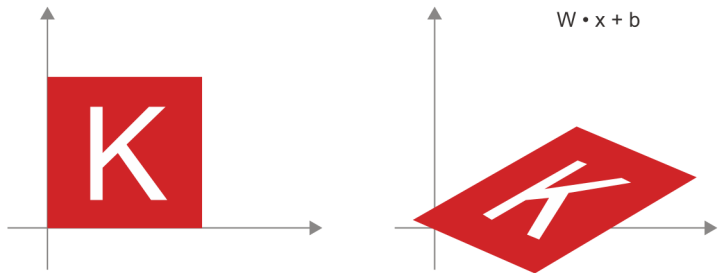


Figure 5: Affine transform in the plane. (Chollet, [2021](#), pg.46)

relu Nonlinear Activation Function

- The previous are just combinations of linear transformations.
- Multilayer NN made entirely of linear transformations can be reduced to a single linear transformation.
- Activation functions produce a nonlinear translation.

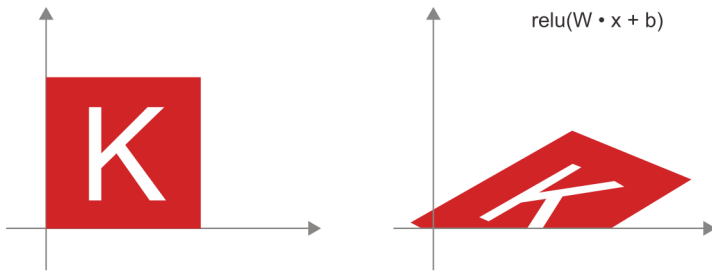


Figure 6: Affine transform followed by (nonlinear) 'relu' activation. (Chollet, [2021](#), pg.47)

A Geometric Interpretation of Deep Learning

- NN consist entirely of chains of tensor operations.
- Can be interpreted as geometric transformations of the input data into a new space/shape/manifold.
- Can interpret a NN as a very complex geometric transformation in a high-dimensional space, implemented via a series of simple linear + nonlinear transformations.



Figure 7: Uncrumpling a complicated manifold of data. (Chollet, [2021](#), pg.47)



L02.4 The Engine of Neural Networks: Gradient-based Optimization

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L02.5 Looking Back at our First Neural Network Example

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Chollet, F. (2021). *Deep learning with python* (second). Manning.